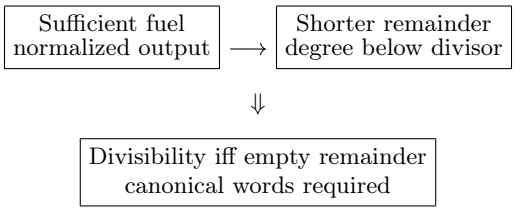
Monic remainder: exact divisibility criterion

T-1198; 2026-10-31. Producer proof packet. Independent source-fidelity review pending.



Derived result. With fuel at least the original input coefficient-list length, the checked monic remainder returns Some(empty) if and only if the divisor polynomial divides the input polynomial. Assumptions are prime 0<p<2^31, canonical divisor/input words and divisor final word equal to one. This combines the preceding termination and successful-return soundness packets with completeness of the empty-result test.

Construction. Every returned list remains normalized: empty, or with final word nonzero. A canonical normalized list represents the zero polynomial exactly when it is empty. A list's polynomial degree is strictly less than its length; a divisor ending in one has a nonzero coefficient at length minus one. Thus a shorter remainder has degree below the divisor. If that remainder is divisible by the divisor, it must be the zero polynomial and therefore the empty list. The existing loop invariant transports divisibility between input and remainder.

Checks. Ten main declarations, eight controls and 18 separate axiom readbacks passed with unchanged input bindings. Controls cover divisible and nondivisible inputs, initial trailing zeros, zero polynomial input, nonmonic rejection, insufficient fuel, a non-normal zero list and a noncanonical zero counterexample. The list [17] modulo 17 is nonempty and normalized as words, yet its polynomial is zero. This explicitly records why the canonical-word hypothesis cannot be dropped. No sampled experiment or runtime measurement is involved.

Native connection. Jacobian.valid in src/cryptanalysis/three_leads.rs checks degree at most two, top coefficient one, and an empty prem of the residual f-v^2 by u. This packet supplies the exact-divisibility criterion for the checked remainder model. Canonicality and polynomial decoding of the caller's residual construction must still be linked to native psub/pmul. The model excludes quotient writes and instrumentation, and uses the odd prime-field interpretation with diagnostic tables disabled. Formal compiled Rust equivalence and independent source-fidelity review remain separate obligations. Existing imported compiled dependencies retain their recorded trust boundary.

Retained failures. The first control check needed explicit unfolding of the Normal predicate for decidability and an explicit finite-field equality for the noncanonical counterexample. A second control attempt exposed a modulus-type coercion mismatch; using the exact modulus self-cast identity fixed it. Corrected controls passed; both failed receipts remain. Main theorem checks passed on their first attempts.

Sources. Native three_leads.rs pdiv/prem and Jacobian.valid; T-1194 successful-return soundness; T-1196 strict descent and sufficient fuel; pinned Mathlib Polynomial/Degree/Defs.lean and Domain.lean (degree_lt_iff_coeff_zero and eq_zero_of_dvd_of_degree_lt); official Mathlib monic-division reference, modByMonic_eq_zero_iff_dvd, inspected 2026-10-31: https://leanprover-community.github.io/mathlib4_docs/Mathlib/Algebra/Polynomial/Div.html. The proof implements a standard degree argument for the existing word/list model; no novelty claim is made.

Next construction. Bind canonical residual construction f-v^2 to the native multiplication/subtraction routines and compose the complete divisor-validity acceptance statement. This packet does not identify every nonzero output with Mathlib's canonical remainder. The full scientific cohort and execution prerequisites remain unchanged; no scientific workers were launched.